

ESERCIZIO

Sia X una v.a. con densità $f_X(x) = c \times 1_{(4,9)}(x)$.

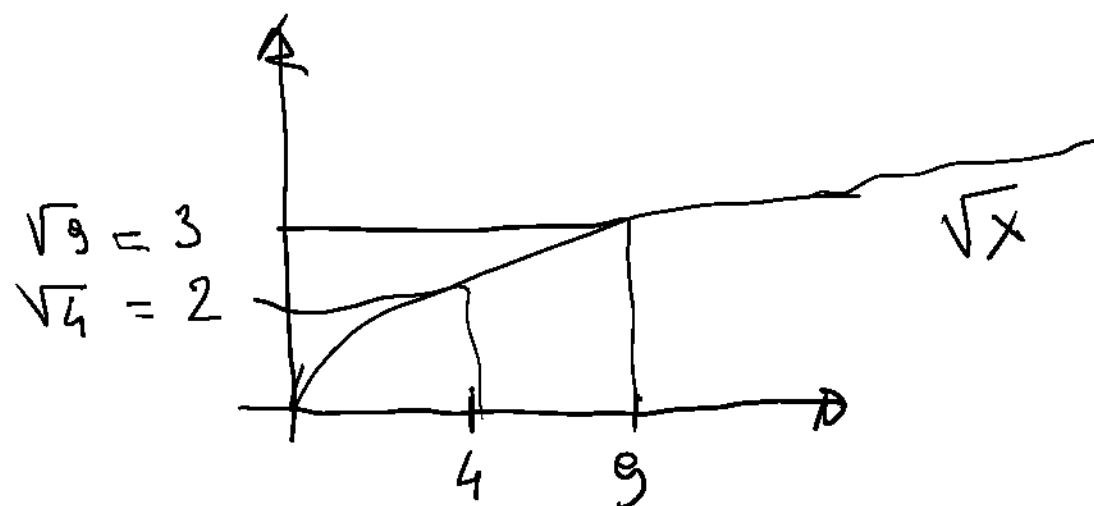
$$\begin{array}{r} 81 \\ 16 \\ \hline 65 \end{array}$$

- 1) Trovare il valore di c
- 2) Trovare la funzione di distribuzione di $Y = \sqrt{X}$
- 3) Calcolare $E[Y]$.

SOLUZIONE

$$\begin{aligned} 1) \quad 1 &= \int_{-\infty}^{\infty} f_X(x) dx = c \int_4^9 x dx = c \left[\frac{x^2}{2} \right]_{x=4}^{x=9} = \frac{c}{2} (9^2 - 4^2) = \frac{c}{2} (81 - 16) \\ &= c \frac{65}{2} \longrightarrow c = \frac{2}{65} \end{aligned}$$

2)



$$P(Y \in [2, 3]) = 1$$

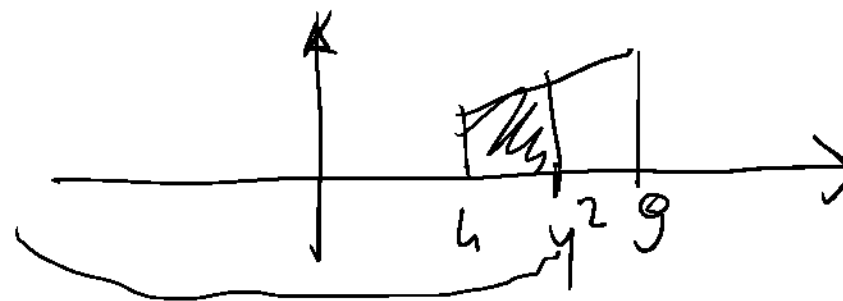
$$F_Y(y) = \begin{cases} 0 & \text{per } y < 2 \\ \textcircled{*} & \text{per } 2 < y < 3 \\ 1 & \text{per } y \geq 3 \end{cases}$$

per $y < 2$
 per $2 < y < 3$
 per $y \geq 3$

$$\textcircled{*} = P(Y \leq y) = P(\sqrt{X} \leq y) = P(X \leq y^2)$$

$$= \int_4^{y^2} \frac{2}{65} x \, dx = \frac{2}{65} \left[\frac{x^2}{2} \right]_{x=4}^{x=y^2} = \frac{y^4 - 16}{65}$$

$$4 < y^2 < 9 \text{ perché}$$



~~3) 1/65~~

3) 1^{er} mode

$$\frac{65}{5} = 13$$

$$E[Y] = E[\sqrt{X}] = \int_4^9 \sqrt{x} \left(\frac{2}{65} \right) x \, dx = \frac{2}{65} \int_4^9 \underbrace{x \sqrt{x}}_{=x \cdot x^{1/2}} \, dx =$$

$$= \frac{2}{65} \int_4^9 x^{3/2} \, dx = \frac{2}{65} \left[\frac{x^{5/2}}{\frac{5}{2}} \right]_{x=4}^{x=9} = \frac{2}{65} \cdot \frac{2}{5} (9^{5/2} - 4^{5/2}) =$$

$$= \frac{4}{325} (3^5 - 2^5) = \frac{4}{325} (243 - 32) = \frac{4}{325} \cdot 181 = \frac{724}{325}$$

$$\begin{array}{r} 181 \\ 4 \\ \hline 724 \end{array}$$

2° modo

$$E[Y] = \int_2^3 y f_Y(y) dy = \int_2^3 y \underbrace{\frac{4y^3}{65}}_{=f_Y(y)} dy =$$

no derivata di $\frac{y^4-16}{65}$

$$= \int_2^3 \frac{4}{65} y^4 dy = \frac{4}{65} \int_2^3 y^4 dy =$$

$$= \frac{4}{65} \left[\frac{y^5}{5} \right]_{y=2}^{y=3} = \frac{4}{\underbrace{65 \cdot 5}_{=325}} [3^5 - 2^5] = \frac{4}{325} \cdot (243 - 32) = \frac{4}{325} \cdot 181 = \frac{724}{325}$$